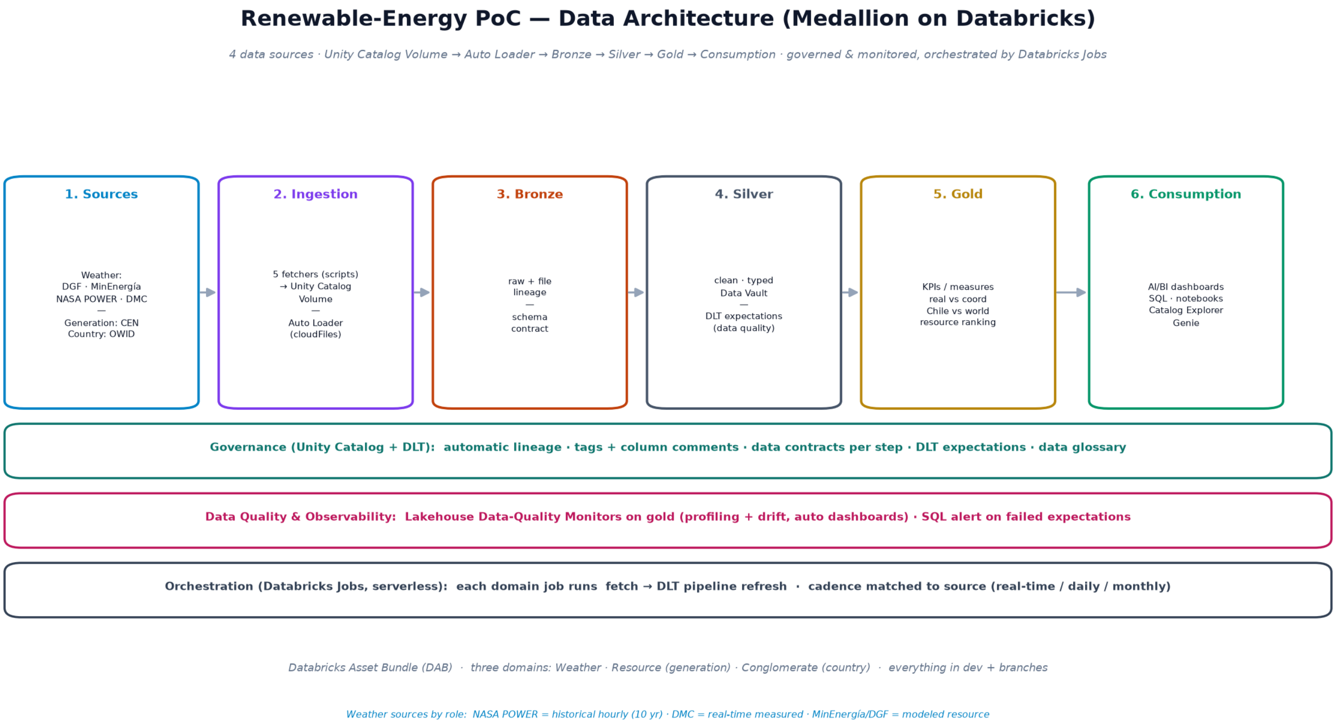
Renewable-Energy PoC - Technical Documentation

Medallion lakehouse on Databricks | updated 2026-07-17

1. Architecture

Databricks Asset Bundle with DLT on serverless, Auto Loader (cloudFiles) ingestion from Unity Catalog Volumes, UC governance. Sources -> Ingestion -> Bronze -> Silver -> Gold -> Consumption, orchestrated by Jobs and monitored with Lakehouse Data-Quality Monitors.



2. Domains & sources

- Weather: NASA POWER (historical hourly) - DMC (real-time measured) - MinEnergia/DGF (modeled).
- Resource: CEN/Coordinador generation (coordinado/real/reducciones); silver = Data Vault.
- Conglomerate: OWID country stats (Entity/Year).

3. Data acquisition (fetchers)

- nasa_power_fetcher.py - NASA POWER hourly by lat/lon, yearly chunks, DMC-equivalent fields.
- cen_coordinador_fetcher.py - CEN SIPUB v1 (public key), generation per plant.
- owid_conglomerate_fetcher.py - OWID 5 country datasets.
- dgf_explorador_fetcher.py / minenergia_api_fetcher.py - modeled/official weather.
- DMC getDatosRecientesEma (usuario+token) - real-time measured stations.

4. Weather - NASA POWER historical load

- Params: GHI (ALLSKY_SFC_SW_DWN), T2M, RH2M, PS, PRECTOTCORR, WS10M/WD10M/WS50M.
- Full load: 876,720 rows (10 plants x 87,672 hours, 2015-2024) -> bronze_weather_nasa_power.
- Physically validated: midday GHI highest for Atacama plants (~871 W/m2).

5. Governance, DQ & observability

- Contracts per step (docs/data-contracts.md) + DLT expectations on silver + UC constraints.
- UC lineage + column comments + data glossary.
- Three Data-Quality Monitors on gold (profiling + drift, auto dashboards) + SQL alert on failures.

6. Deploy & run (dev only)

databricks bundle deploy -t dev --var=dev_catalog=workspace

databricks bundle run {weather_streaming_job | conglomerate_owid_job | cen_generation_job} -t dev --var=dev_catalog=workspace

All work is in dev and on branches. Prod is managed by the project lead.

Verify the medallion & jobs

- Resource pipeline: https://dbc-54b27bae-2e91.cloud.databricks.com/pipelines/9bcfc32a-19f0-4327-b5bc-e8ee1cc4a3a4?o=7474645896934260
- Conglomerate pipeline: https://dbc-54b27bae-2e91.cloud.databricks.com/pipelines/b5d63282-a439-42ef-8e3b-cd707a9d5f58?o=7474645896934260
- Weather pipeline: https://dbc-54b27bae-2e91.cloud.databricks.com/pipelines/c108b287-98cc-43d6-86d6-5b63a9e6b4ae?o=7474645896934260
- Weather job: https://dbc-54b27bae-2e91.cloud.databricks.com/jobs/503330163541320?o=7474645896934260
- Quality monitor (Resource gold): https://dbc-54b27bae-2e91.cloud.databricks.com/sql/dashboardsv3/01f1809adb1a199384b80e96c469a206?o=7474645896934260
- Pull requests: https://github.com/oxiboy/poc_databricks_evalueserve/pulls